METHODS DEVELOPMENT NOTE / ROUGH DRAFT

Extending the amyloid mixture-cure ODE model to longitudinal tau PET

Core model, dependence assumptions, and MCMC implementation

Status. This is a proposed extension of the supplied amyloid-only R implementation, not a fitted joint model or a finalized manuscript section. It records the agreed three-state susceptibility restriction and a concrete first dependence structure. Tau-specific calibration and implementation choices remain open.

1 Observed data and subject-level quantities

Let $i = 1, \ldots, N$ index subjects and $k \in \{A, T\}$ index amyloid and tau. Let $j = 1, \ldots, n_{ki}$ index that subject's scans of biomarker k. Define

$$a_{ij}^k = \text{calendar age at scan}, \qquad y_{ij}^k = \text{observed PET value}, \qquad D_i^k = \{(a_{ij}^k, y_{ij}^k) : j = 1, \ldots, n_{ki}\}.$$

Uppercase Y_{ij}^k denotes a random scan value; lowercase y_{ij}^k is its observed realization. The complete data are $D = \{D_i^A, D_i^T\}_{i=1}^N$. Ages and observed covariates are conditioned on throughout; their densities are not included in the likelihood. Visits need not be matched across biomarkers. If $n_{ki} = 0$, its observation-likelihood contribution is one. Treating absent scans this way assumes an ignorable observation process.

Symbol	Definition
$Z_{ki} \in \{0, 1\}$	Whether subject i is eventually an accumulator of biomarker k. This is not current PET positivity.
$S_i = (Z_{Ai}, Z_{Ti})$	Subject's susceptibility state; allowed values are 00, 10, and 11.
α_{ki}	Departure age: when an accumulator leaves the common launch level.
δ_{ki}	Log speed multiplier; $\exp(\delta_{ki})$ rescales progression time.
b_{ki}	Subject's constant latent level in the non-accumulator branch.
μ_{bk}	Biomarker-specific common launch level for accumulators and mean parameter for non-accumulator levels.
$M_i^k(a)$	Subject's noise-free biomarker level at calendar age a, in the active mixture branch.
η_i	Continuous latent vector $(\alpha_{Ai}, \alpha_{Ti}, \delta_{Ai}, \delta_{Ti}, b_{Ai}, b_{Ti})$.
Θ	All population parameters: spline coefficients, regression coefficients, scale and smoothing parameters, and dependence parameters. Fixed settings are excluded.

Interpretation of inactive variables. Departure age and speed have biological meaning only for an accumulator. A constant non-accumulator level is active only in a non-accumulator branch. Section 4 explains how proper auxiliary distributions let the sampler retain a fixed-dimensional latent vector for every subject.

2 Retain two separate clocks and observation models

For each biomarker k, let c_k be its positivity threshold and let $[\ell_k, h_k]$ be its computational level grid. Let $B_k(y)$ and $V_k(y)$ be fixed spline basis vectors of lengths K_k and Q_k. Their coefficient vectors are θ_k and ν_k, respectively. Define the population rate and measurement-scale functions by

$$r_k(y) = \exp\{B_k(y)^\top \theta_k\}, \qquad \omega_k(y) = \exp\{V_k(y)^\top \nu_k\}.$$

Let $r_{\min,k} > 0$ be a numerical rate floor and write $\bar{r}_k(y) = \max\{r_k(y), r_{\min,k}\}$. The corresponding clock is

$$G_k(y) = \int_{c_k}^{y} \frac{du}{\bar{r}_k(u)}. \tag{1}$$

It is strictly increasing on its domain. Its inverse is denoted by G_k^{-1}. Both the integral and its inverse are evaluated numerically on a grid, as in the current model; the clock transformation does not require a repeated time-stepping ODE solve.

For an accumulator, the latent trajectory is

$$M_i^{k,(1)}(a) = G_k^{-1}\left[G_k(\mu_{bk}) + e^{\delta_{ki}}(a - \alpha_{ki})_+\right], \qquad (x)_+ = \max(x, 0). \tag{2}$$

It equals μ_{bk} before departure and follows $dM/da = e^{\delta_{ki}}\bar{r}_k(M)$ afterward. Where the numerical floor is inactive, this is the original ODE with rate r_k. For a non-accumulator, $M_i^{k,(0)}(a) = b_{ki}$.

Let $\sigma_{bk} > 0$ be the between-subject non-accumulator scale. Retain the current family

$$b_{ki} \sim \mathcal{N}(\mu_{bk}, \sigma_{bk}^2) \text{ truncated to } b_{ki} < c_k.$$

The truncation normalizer is part of this density. A noisy observed PET value may exceed c_k even when its latent non-accumulator level is below c_k.

Let d_k be the fixed Student-t degrees of freedom, initially $d_A = d_T = 4$. Define $t_d(y; m, s)$ as the Student-t density at y with location m and scale $s > 0$. For either branch $z \in \{0, 1\}$,

$$Y_{ij}^k \mid \eta_i, Z_{ki} = z, \Theta \sim t_{d_k}\left(M_i^{k,(z)}(a_{ij}^k), \omega_k(M_i^{k,(z)}(a_{ij}^k))\right). \tag{3}$$

Here ω_k is a *scale*, not a standard deviation: the conditional variance is $d_k\omega_k^2/(d_k - 2)$ when $d_k > 2$.

What remains separate. Amyloid and tau get their own $\theta_k, \nu_k, \mu_{bk}, \sigma_{bk}$, grids, thresholds, and spline bases. Tau values must be calibrated to their own tracer, processing pipeline, and chosen region of interest. This draft treats tau as one scalar longitudinal PET summary, not a spatial field.

3 Three susceptibility states and conditional independence

The simplifying structural assumption is $Z_{Ti} \leq Z_{Ai}$. It rules out tau-only accumulators. It does not impose departure-age ordering or threshold-crossing ordering.

Let w_i and v_i be known covariate vectors, including intercepts, for amyloid susceptibility and conditional tau susceptibility. Let ϕ_A and $\phi_{T|A}$ be their regression coefficient vectors. Define $\text{expit}(x) = (1 + e^{-x})^{-1}$ and

$$\pi_i = \text{expit}(w_i^\top \phi_A), \qquad q_i = \text{expit}(v_i^\top \phi_{T|A}).$$

Then $\pi_i = P(Z_{Ai} = 1 \mid \Theta)$ and $q_i = P(Z_{Ti} = 1 \mid Z_{Ai} = 1, \Theta)$. The state probabilities, written $w_{si} = P(S_i = s \mid \Theta)$, are

$$\begin{aligned}
s = 00: &\quad w_{00,i} = 1 - \pi_i, &&\text{neither ever accumulates,}\\
s = 10: &\quad w_{10,i} = \pi_i(1 - q_i), &&\text{amyloid only,}\\
s = 11: &\quad w_{11,i} = \pi_i q_i, &&\text{both eventually accumulate.}
\end{aligned}$$

They sum to one. There is no separate susceptibility-correlation parameter.

Observation-level assumption. Given S_i, η_i, and Θ, assume scan errors are independent across biomarkers and visits. Consequently,

$$p(D_i^A, D_i^T \mid \eta_i, S_i, \Theta) = p(D_i^A \mid \eta_i, S_i, \Theta)\, p(D_i^T \mid \eta_i, S_i, \Theta). \tag{4}$$

This is an assumption about the sampling distribution, like independent errors in regression after the fitted means have been specified. It does not follow automatically from writing two ODEs.

Define the conditional branch likelihood, evaluated at the observed scans, as

$$L_{ki}^{(z)}(\eta_i, \Theta) = \prod_{j=1}^{n_{ki}} t_{d_k}\left(y_{ij}^k; M_i^{k,(z)}(a_{ij}^k), \omega_k(M_i^{k,(z)}(a_{ij}^k))\right). \tag{5}$$

These likelihood factors are conditional on proposed latent values. They have not already been integrated over departure ages or speeds.

Dependence remains. The susceptibility states are dependent by construction. Section 4 additionally links departure ages and speeds. After averaging over those unknown quantities, the amyloid and tau records are generally dependent. For example,

$$P(Z_{Ai} = 1, Z_{Ti} = 1) = \pi_i q_i \neq \pi_i^2 q_i = P(Z_{Ai} = 1)P(Z_{Ti} = 1)$$

except at special parameter values. Residual dependence beyond the latent trajectories could be added later by replacing the product in (4) with a joint observation model.

4 Connect departure ages and speeds through joint distributions

Let x_{ki} be a known departure-age covariate vector including an intercept, and u_{ki} a centered speed covariate vector without an intercept. Let γ_k and ψ_k be their respective coefficients. Let $a_{\min,k} < a_{\max,k}$ be fixed departure-age bounds, $\sigma_{\alpha k} > 0$ a Weibull shape-control parameter, and $\sigma_{\delta k} > 0$ a speed scale.

Departure-age marginals. Set $\lambda_{ki} = \exp(x_{ki}^\top \gamma_k)$ and $\kappa_k = 1/\sigma_{\alpha k}$. For $a_{\min,k} < a < a_{\max,k}$, define the shifted, truncated Weibull CDF

$$F_{ki}(a) = \frac{1 - \exp\{-[(a - a_{\min,k})/\lambda_{ki}]^{\kappa_k}\}}{1 - \exp\{-[(a_{\max,k} - a_{\min,k})/\lambda_{ki}]^{\kappa_k}\}}. \tag{6}$$

Extend this CDF by zero below the lower bound and one above the upper bound. Its density is $f_{ki}(a) = F'_{ki}(a)$ in the interior. This preserves the current marginal family.

Speed marginals. Write $\delta_{ki} = u_{ki}^\top \psi_k + \sigma_{\delta k} z_{ki}$. Retain $z_{ki} \sim \mathcal{N}(0,1)$ truncated to $[-c_z, c_z]$, with fixed $c_z = 4$ initially. Let H be this truncated-normal CDF and h its density. The CDF and density of δ_{ki} are

$$R_{ki}(d) = H\left(\frac{d - u_{ki}^\top \psi_k}{\sigma_{\delta k}}\right), \qquad g_{ki}(d) = \frac{1}{\sigma_{\delta k}} h\left(\frac{d - u_{ki}^\top \psi_k}{\sigma_{\delta k}}\right).$$

Dependence with preserved marginals. Let Φ and φ be the standard-normal CDF and density, and $\varphi_2(\cdot, \cdot; \rho)$ the bivariate standard-normal density with correlation $\rho \in (-1, 1)$. For probability arguments $p_1, p_2 \in (0, 1)$, its Gaussian copula density is

$$c_\rho(p_1, p_2) = \frac{\varphi_2(\Phi^{-1}(p_1), \Phi^{-1}(p_2); \rho)}{\varphi(\Phi^{-1}(p_1))\varphi(\Phi^{-1}(p_2))}.$$

Use separate parameters ρ_α and ρ_δ. For generic ages (a_A, a_T) and log speeds (ξ_A, ξ_T), define

$$f_{\alpha i}(a_A, a_T) = c_{\rho_\alpha}(F_{Ai}(a_A), F_{Ti}(a_T)) f_{Ai}(a_A) f_{Ti}(a_T), \tag{7}$$
$$f_{\delta i}(\xi_A, \xi_T) = c_{\rho_\delta}(R_{Ai}(\xi_A), R_{Ti}(\xi_T)) g_{Ai}(\xi_A) g_{Ti}(\xi_T). \tag{8}$$

These correlations act on latent normal scores; they are not generally Pearson correlations of the original ages or speeds. Initially take the departure-age pair independent of the speed pair, conditional on covariates and Θ.

Common augmentation. Draw η_i from these proper joint distributions and independent baseline-level densities, independently of S_i given Θ and covariates. Inactive coordinates are auxiliary. This preserves the same amyloid marginal history in states 10 and 11; in state 10 the unused tau history integrates out. State-specific history distributions are a possible later extension, but would change the mixture construction.

5 The joint likelihood, posterior, and information sharing

Let f_{bk} denote the normalized truncated-normal density of b_{ki} from Section 2. Under the common augmentation, the joint latent density is

$$f_i(\eta_i \mid \Theta) = f_{\alpha i}(\alpha_{Ai}, \alpha_{Ti})\, f_{\delta i}(\delta_{Ai}, \delta_{Ti})\, f_{bA}(b_{Ai}) f_{bT}(b_{Ti}). \tag{9}$$

It does not depend on S_i. This explicit assumption makes the following state-marginalized *conditional observation likelihood* valid:

$$\begin{aligned}\mathcal{L}_i(\eta_i, \Theta) = (1-\pi_i) L_{Ai}^{(0)} L_{Ti}^{(0)} \\ + \pi_i (1-q_i) L_{Ai}^{(1)} L_{Ti}^{(0)} + \pi_i q_i L_{Ai}^{(1)} L_{Ti}^{(1)}.\end{aligned} \tag{10}$$

The arguments (η_i, Θ) of the branch likelihoods are suppressed only in this display. The observation factors multiply because of (4); f_i is included separately in the posterior, not omitted from the model.

Let $p(\Theta)$ denote the population prior. Assuming subjects are independent conditional on population parameters, the posterior target is

$$p(\Theta, \eta_1, \ldots, \eta_N \mid D) \;\propto\; p(\Theta) \prod_{i=1}^{N} \mathcal{L}_i(\eta_i, \Theta) f_i(\eta_i \mid \Theta). \tag{11}$$

The likelihood for population parameters alone is instead

$$p(D \mid \Theta) = \prod_{i=1}^{N} \int \mathcal{L}_i(\eta_i, \Theta) f_i(\eta_i \mid \Theta)\, d\eta_i. \tag{12}$$

MCMC need not numerically evaluate all these integrals. Sampling the continuous latent coordinates and averaging posterior draws performs the required integration implicitly.

State probabilities. Let logsumexp denote a numerically stable log-sum-exp. For a state $s = (z_A, z_T)$ with components $z_A, z_T \in \{0, 1\}$, define

$$\ell_{si} = \log w_{si} + \log L_{Ai}^{(z_A)} + \log L_{Ti}^{(z_T)}, \qquad r_{si} = \exp\{\ell_{si} - \operatorname{logsumexp}_{t \in \{00,10,11\}} \ell_{ti}\}.$$

At a posterior draw, r_{si} is $P(S_i = s \mid D_i^A, D_i^T, \eta_i, \Theta)$. Averaging r_{si} over draws yields the posterior state probability. The three probabilities sum to one at every draw. Biological history summaries should condition on susceptibility. For example, with $\mathbb{E}$ denoting posterior expectation,

$$\mathbb{E}(\alpha_{Ti} \mid Z_{Ti} = 1, D) = \frac{\mathbb{E}(r_{11,i} \alpha_{Ti} \mid D)}{\mathbb{E}(r_{11,i} \mid D)}.$$

Intuition. A high tau scan can increase the weight of state 11, and thereby amyloid susceptibility. A proposed amyloid departure age changes its joint density with the tau departure age, so tau data can constrain amyloid timing. A speed update behaves similarly. Independent scan errors allow the observation factors to multiply; correlated latent histories and shared state weights connect the inverse problem.

6 Keep the current location-coordinate reparameterization

The current sampler uses a subject's location at a reference age rather than sampling departure age directly. For each biomarker let $\widetilde{a}_{ki}$ be the mean of its scan ages, or a fixed reference age when that biomarker is unobserved. Extend the clock below the launch level:

$$G_{k,\mathrm{ext}}(x) = \begin{cases} G_k(x), & x \geq \mu_{bk}, \\ G_k(\mu_{bk}) + (x - \mu_{bk})/\bar{r}_k(\mu_{bk}), & x < \mu_{bk}. \end{cases}$$

Let $\widetilde{x}_{ki}$ be the location coordinate. Its implied departure age is

$$\alpha_{ki} = \widetilde{a}_{ki} - e^{-\delta_{ki}}\{G_{k,\mathrm{ext}}(\widetilde{x}_{ki}) - G_k(\mu_{bk})\}. \tag{13}$$

If $\widetilde{x}_{ki} < \mu_{bk}$, it is a coordinate encoding future departure, not an actual biomarker level below the common launch level. This preserves a continuous coordinate for subjects who have not departed at their reference age.

For fixed δ_{ki} and population parameters, define the positive absolute Jacobian

$$J_{ki} = \left|\frac{\partial \alpha_{ki}}{\partial \widetilde{x}_{ki}}\right| = \begin{cases} e^{-\delta_{ki}}/\bar{r}_k(\widetilde{x}_{ki}), & \widetilde{x}_{ki} \geq \mu_{bk}, \\ e^{-\delta_{ki}}/\bar{r}_k(\mu_{bk}), & \widetilde{x}_{ki} < \mu_{bk}. \end{cases}$$

The joint location-coordinate density, conditional on the speeds, is therefore

$$p(\widetilde{x}_{Ai}, \widetilde{x}_{Ti} \mid \delta_{Ai}, \delta_{Ti}, \Theta) = f_{\alpha i}(\alpha_{Ai}, \alpha_{Ti}) J_{Ai} J_{Ti}. \tag{14}$$

The Jacobian determinant remains a product: each departure-age transformation depends on its own location coordinate when the speeds are held fixed.

Practical modification of `dxtilde`. The two existing-style marginal densities supply $f_{Ai}J_{Ai}$ and $f_{Ti}J_{Ti}$. Add

$$\log c_{\rho_\alpha}\{F_{Ai}(\alpha_{Ai}), F_{Ti}(\alpha_{Ti})\}$$

exactly once to obtain the joint density. Do not also add a second full Weibull prior. A single bivariate custom density is a clean way to collect these terms. Likewise, in non-centered speed coordinates (z_{Ai}, z_{Ti}), use $h(z_{Ai})h(z_{Ti})$ times the speed copula correction. The $1/\sigma_{\delta k}$ factors belong to densities in δ coordinates, not to a prior directly on z.

Numerical support. Retain explicit feasibility checks and stable tail calculations. A finite grid must support all evaluated trajectories. Extending a clock numerically and imposing a statistical support restriction are different operations; any parameter-dependent truncation requires its proper normalizer. Below-floor location coordinates must not be clipped to the physical level-grid minimum.

7 How MCMC fits the components jointly

Use (11) expressed in location and non-centered speed coordinates. Population parameters remain in Θ; the sampled subject coordinates are now $(\widetilde{x}_{Ai}, z_{Ai}, b_{Ai}, \widetilde{x}_{Ti}, z_{Ti}, b_{Ti})$. Speeds and departure ages are deterministic functions of these coordinates and Θ.

1. **Build both population clocks.** Given the current spline coefficients, compute r_A, r_T, their clock grids, and both measurement-scale grids. Recompute affected grids when their coefficients change.

2. **Evaluate each subject's three branches.** Construct both latent trajectories at their own scan ages. Evaluate $\ell_{00,i}, \ell_{10,i}, \ell_{11,i}$ and their log-sum-exp. The discrete state is summed out, so the chain need not sample Z_{Ai} or Z_{Ti} explicitly.

3. **Update subject histories.** A first proposal is a joint random-walk block for $(\widetilde{x}_{Ai}, z_{Ai}, \widetilde{x}_{Ti}, z_{Ti})$. It addresses within-biomarker timing–speed trade-offs and cross-biomarker dependence. Update b_{Ai}, b_{Ti} separately initially. Compare this four-coordinate block with smaller paired blocks using effective samples per unit time; larger blocks are not automatically more efficient.

4. **Update population parameters.** Start with biomarker-specific blocks analogous to the current sampler: departure-age intercept with rate coefficients, measurement coefficients together, other covariate effects in their own blocks, and scalar scales/floors with suitable slice updates. Every update must include all affected observation, copula, marginal-density, and Jacobian terms.

5. **Update dependence and susceptibility.** Updating ρ_α changes joint timing densities; updating ρ_δ changes joint speed densities. Updating ϕ_A or $\phi_{T|A}$ changes the three mixture weights across subjects. All of these steps target the same posterior.

A concrete update. If a proposal moves an amyloid departure age later, its own scans may fit worse unless speed also rises. Its compatibility with the tau departure age also changes through $f_{\alpha i}$. The Metropolis ratio, or the density evaluated by a slice sampler, includes both consequences. The sampler therefore does not fit amyloid first and hold it fixed while fitting tau.

Three different notions of independence. Conditional observation independence is an assumption in Section 3. Conditional independence between the timing pair and speed pair is a separate simplifying latent-distribution assumption. Neither implies posterior independence: the likelihood can induce posterior dependence among all these quantities. A sampler may update blocks sequentially while still targeting their full joint posterior.

Auxiliary variables. Coordinates inactive in a mixture branch remain mathematically useful but can slow mixing. They should not be reported as biological estimates for non-accumulators. Marginalizing a state while retaining a common latent vector avoids changing parameter dimension between iterations.

8 Changes to the code, priors, and first validation steps

Current component	Planned change
`00_settings.R`	Add tau-specific grids, positivity threshold, launch/noise scales, and dependence-prior settings. Keep amyloid settings fixed for the baseline extension.
`01_integrator.R`	Reuse generic clock, interpolation, and inverse-clock routines separately for each biomarker.
`02_bases.R`	Parameterize basis builders by biomarker; build rate and noise bases on each biomarker's scale.
`03_densities.R`	Add joint timing/speed density terms; replace the two-branch observation density with the three-state amyloid–tau density.
`04_model.R`	Add tau latent coordinates and population parameters, conditional tau susceptibility regression, two dependence parameters, and their posterior dependencies.
Runner and figures	Configure joint blocks; retain chain identities; report both biomarkers, state probabilities, dependence estimates, and conditional subject histories.

Priors: retain families, calibrate scales. Carry over the adaptive random-walk family for each rate-coefficient vector and the bounded random walk for each log measurement-scale vector. In that rate prior, $\zeta_k > 0$ denotes global precision and $\lambda_{kj}^{\theta} > 0$ a local variance multiplier:

$$\theta_{kj} - \theta_{k,j-1} \mid \lambda_{kj}^{\theta}, \zeta_k \sim \mathcal{N}(0, \lambda_{kj}^{\theta}/\zeta_k).$$

Normal regression priors, positive scale priors, and launch-level truncations can follow the current families. Tau hyperparameters and bounds require calibration on the chosen tau summary; copying amyloid SUVR numbers is not justified. Give ρ_α, ρ_δ proper symmetric priors on $(-1, 1)$; their concentration near zero remains to be chosen. Any pinned spline coefficient must be justified from that biomarker's basis support and identification, rather than copied by index. The speed regression initially has no intercept, preserving a reference scale for the shared rate curve.

Implementation sequence.

1. Fit/check a tau-only analogue first; choose its scalar region/tracer summary and numerical domain.
2. Build the three-state model with $\rho_\alpha = \rho_\delta = 0$; verify branch weights and recovery on simulated data.
3. Add timing dependence, then speed dependence. Check recovery near zero and at nonzero dependence, with realistic asynchronous and missing scans.
4. Check rank-normalized split-$\widehat{R}$, bulk/tail ESS, and Monte Carlo precision for population parameters, state probabilities, and representative subject quantities. Validate independent-start sensitivity and prior sensitivity.

Still open. Departure ordering is not yet imposed. If desired, define a positive tau lag or a properly normalized ordered joint distribution. Residual scan correlation, dependence between timing and speed, correlated flat levels, state-specific progression distributions, and spatial tau dynamics are later extensions. The three-state restriction is a simplifying hypothesis to assess against the intended data, not an empirical conclusion established by this draft.

Source of the baseline: supplied `00_settings.R`–`05_figures.R`, `README.md`, and `run_model(1).R`. Equations for the joint model above are proposals developed from that implementation. No joint amyloid–tau fit or recovery result is claimed.